

Sieve of Eratosthenes

1 Problem

Goal

Given an integer n , find all prime numbers from 2 to n .

Main Idea

Start by assuming every number is prime. Then repeatedly choose the next unmarked number p , and mark all multiples of p as composite.

2 Why the Sieve Works

If a number m is composite, then

$$m = ab$$

for some integers $a, b > 1$.

At least one of a and b is at most $\sqrt{m}$. So every composite number has a prime factor at most its square root.

Stopping Rule

When sieving up to n , it is enough to mark multiples for

$$p \leq \sqrt{n}.$$

3 Algorithm Steps

1. Create a Boolean array `isPrime[0...n]`.
2. Mark 0 and 1 as not prime.
3. Initially mark every number $2, 3, \dots, n$ as prime.
4. For each $p = 2, 3, \dots, \lfloor \sqrt{n} \rfloor$, if p is still marked prime, mark

$$p^2, p^2 + p, p^2 + 2p, \dots$$

as composite.

5. The numbers still marked prime are the primes up to n .

Sieve of Eratosthenes

```
Sieve(n):  
  for i = 0 to n:  
    isPrime[i] = true
```

```

isPrime[0] = false
isPrime[1] = false

for p = 2 to floor(sqrt(n)):
    if isPrime[p] == true:
        for multiple = p*p to n step p:
            isPrime[multiple] = false

return all i such that isPrime[i] == true

```

4 Why Start from p^2 ?

Important

When processing p , multiples smaller than p^2 have already been marked by smaller prime factors.

For example, when $p = 5$, the smaller multiples are

$$10 = 2 \cdot 5, \quad 15 = 3 \cdot 5, \quad 20 = 4 \cdot 5.$$

These were already handled before reaching 5. So the first new multiple is

$$5^2 = 25.$$

5 Worked Example: Primes up to 30

Problem

Find all primes from 2 to 30 using the Sieve of Eratosthenes.

Step 1: Start

Begin with

2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30.

Step 2: Mark Multiples of 2

Keep 2. Mark

4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30

as composite.

Step 3: Mark Multiples of 3

Keep 3. Start from

$$3^2 = 9.$$

Mark

9, 12, 15, 18, 21, 24, 27, 30

as composite.

Step 4: Mark Multiples of 5

Keep 5. Start from

$$5^2 = 25.$$

Mark

25, 30

as composite.

Step 5: Stop

Since

$$\sqrt{30} < 6,$$

we only need to check primes up to 5.

Answer

The primes up to 30 are

2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

6 Running Time

The sieve marks multiples of primes:

$$\frac{n}{2} + \frac{n}{3} + \frac{n}{5} + \cdots.$$

The standard running time is

$$\mathcal{O}(n \log \log n).$$

The array uses

$$\mathcal{O}(n)$$

space.

7 Final Review

Core Facts

The sieve finds all primes up to n .
Only unmarked numbers are used as new primes.
For prime p , start marking from p^2 .
Stop after $p > \sqrt{n}$.
Time: $\mathcal{O}(n \log \log n)$, space: $\mathcal{O}(n)$.